

Local-First RAG Research Companion

Discover, index, retrieve and answer over scientific papers on an 8 GB CPU-only Windows machine

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Code, data and results: github.com/Charles2524/Research_RAG

Literature survey (1 of 2): 2025-2026 journal publications

- Yan et al. (2025), SSRN Electronic Journal. Corrective Retrieval Augmented Generation (CRAG).
- Ke et al. (2025), npj Digital Medicine. RAG for 10 LLMs in assessing medical fitness.
- Vishwanath et al. (2025), BMJ Digital Health & AI. MedMobile: a 3.8B clinical model.
- Shen et al. (2025), BMJ Digital Health & AI. LLM-enhanced framework for systematic reviews.
- Klesel & Wittmann (2025), Business & Information Systems Eng.. Retrieval-Augmented Generation (catchword).
- Yang et al. (2025), npj Health Systems. RAG for generative AI in health care (perspective).
- Zhao P. et al. (2026), Data Science and Engineering. RAG for AI-Generated Content: a survey.
- Spens & Burgess (2026), Nature Communications. Hippocampo-neocortical interaction as compressive RAG.

16 works in total: 13 journal articles and 3 preprints, all published 2025 or 2026, all part of the indexed corpus.

Literature survey (2 of 2): 2026 journals and preprints

- Godoy et al. (2026), BMC Med. Informatics & Decision Making. RAG extraction from mammography reports.
- Liu et al. (2026), Digital Health. Dynamic alignment for heart-failure decision support.
- Almohaimeed et al. (2026), Frontiers in Big Data. Benchmarking RAG LLMs for Arabic noise robustness.
- Kang et al. (2026), J. Medical Internet Research. Health-education agent for gastric cancer.
- Kuster et al. (2026), Frontiers in Neurorobotics. VLM-guided RAG for robot action prediction.
- Fasolino (2026), Qeios (preprint). In RAG We Trust? Robustness under document poisoning.
- Defilippo et al. (2026), Research Square (preprint). RAGtio: evaluating hybrid RAG pipelines.
- Priyad et al. (2026), Research Square (preprint). Agentic RAG for verifiable regulatory compliance.

Corpus discovered with the system itself: OpenAlex, arXiv, Europe PMC and PubMed, de-duplicated across sources.

Comparison of the surveyed papers (1 of 2)

Paper	Domain	Retrieval approach	Generator	Key result
Yan et al. 2025	Open-domain QA	Any retriever + T5-large evaluator; web search fallback	LLaMA-2 / others	Retrieval evaluator (T5-large) grades retrieved docs and triggers correct / ambiguous / incorrect actions
Ke et al. 2025	Perioperative medicine	LlamaIndex auto-merging, top-k 30, 58 guidelines	GPT-3.5/4/4o, Gemini, Llama 2/3, Claude	3,234 responses vs 448 human answers; GPT-4 + RAG answered in ~20 s at human-level accuracy
Vishwanath et al. 2025	Medical QA on device	RAG tried as an add-on	phi-3-mini 3.8B (fine-tuned)	75.7% on MedQA; RAG lowered accuracy by 12.6 points on this small model (negative result)
Shen et al. 2025	Evidence synthesis	RAG + agent architecture (framework)	Various	21 studies reviewed; RAG and agents proposed against hallucination and bias; consultant vs assistant roles
Klesel & Wittmann 2025	Enterprise search	Chunk-level retrieval; hierarchical methods discussed	Generic LLM	Names the 'blinkered chunk effect': an isolated chunk cannot be understood without wider document context
Yang et al. 2025	Health care	External-knowledge retrieval (conceptual)	Generic LLM	Frames RAG's contribution as equity, reliability and personalisation; lists implementation limits
Zhao P. et al. 2026	Cross-modal survey	Taxonomy of RAG foundations and enhancements	Any generator	Surveys RAG foundations, enhancements, applications and benchmarks across modalities
Spens & Burgess 2026	Computational neuroscience	Hippocampal episodic retrieval into working memory	Neocortical generative net	Models memory as compressive RAG: episodes compressed, consolidated, retrieved as a basis for generation

Comparison of the surveyed papers (2 of 2)

Paper	Domain	Retrieval approach	Generator	Key result
Godoy et al. 2026	Radiology NLP (Spanish)	all-MiniLM-L6-v2 embeddings; annotated reports as demonstrations	GPT-based + local open-weight	NER F1 0.89-0.94 vs fine-tuned BETO 0.97 with no task training; RE up to 0.78 F1; cents per report
Liu et al. 2026	Cardiology	Guideline RAG as the final alignment stage	LLaMA-3.1 (CPT, SFT, GRPO, RAG)	Clinical score 0.716 -> 0.864 and risk safety 0.820 -> 0.948 with RAG; BLEU-4 fell (alignment tax)
Almohaimeed et al. 2026	Arabic QA	Retrieved documents with injected noise; Gemini-2.5 relevance labels	Several LLMs	Noise-robustness benchmark; relevance of ~300-word documents labelled automatically, 500 checked by hand
Kang et al. 2026	Patient education	Sparse retrieval of 50-100 candidates, medical reranker picks top 3	OpenMEDLab 2.0	Action-research deployment at a cancer centre; accuracy, usability and experience assessed with patients
Kuster et al. 2026	Robotics	Vision-language retrieval over demonstrations	SigLIP2 / CLIP	SigLIP2 reached 68.47% top-1 on cropped images vs 46.88% for CLIP-Large
Fasolino 2026	Fact checking (FEVER)	MiniLM + FAISS, k = 3	Llama 3.1 8B Q4_K_M on CPU (llama.cpp)	Accuracy 77.9% clean -> 43.5% fully poisoned; entity swap flips most answers (23.5% fooled rate)
Defilippo et al. 2026	Biomedical corpora	Sparse, dense, hybrid (RRF 0.7/0.3, k = 60), + cross-encoder rerank	Pipeline-agnostic	Synthetic queries generated from indexed chunks inflate retrieval scores; human queries (Mode B) are stricter
Priyad et al. 2026	Urban cyber-physical systems	Retrieval agent + deterministic critic agent	LLM after verification	nDCG@5 0.75 vs 0.58 for CRAG; supersession detection 0.88; critic P = R = F1 = 1.00 on 78 probes

Research gap from the literature

What the literature measures

- Answer quality on benchmarks, mostly with API-hosted models (GPT-4, Gemini, Claude) or 8B+ open models on GPUs.
- Component novelty: evaluators (CRAG), critics (Priyad), structure-aware retrieval (AeroSAR), graph communities (CGS-RAG).
- Robustness to noise and poisoning (Fasolino; Almohaimeed) for one fixed pipeline.

What it leaves open

- Which retrieval and generation choices matter when the model is small and the machine is an 8 GB CPU laptop: MedMobile shows RAG can even hurt a 3.8B model.
- Systems cost alongside quality: latency, tokens/s, resident memory and index size are rarely reported together (RAGtio is the exception, retrieval only).
- A hard privacy boundary: no paper separates a network discovery layer from an offline pipeline and enforces it by test.
- Citation integrity as a first-class metric: hallucinated references are measured post hoc, not checked against the retrieved set on every answer.
- Evaluation on synthetic questions inflates results (RAGtio); ground truth needs human verification.

Novelty points of this work

Engineering contribution, not a new algorithm

- Local-first by construction: discovery and fetch are the only networked modules; parse, chunk, index, retrieve and generate must not import an HTTP client, enforced by an AST test.
- One-flag offline mode disables every network client; the full pipeline was run under it as a gate.
- Citation-forced prompt plus a post-generation integrity check: every cited id must exist in the retrieved set; violations are counted, never dropped.

Measured, one axis at a time

- Ablation over chunk size, retrieval mode, reranking, context budget and model, each recorded with quality and systems metrics in a single CSV.
- Chunk-level recall (the retrieved chunk must contain the source passage) because paper-level recall saturates on a small corpus.
- Runs on 8 GB CPU-only Windows: sqlite-vec with a NumPy fallback, bge-small embeddings, Qwen3 1.7B/4B via Ollama, peak ~4.1 GB.
- A demo UI that makes verification a click: citation chip -> exact passage, verification line under every answer.

Proposed methodology: producer-consumer architecture

Discovery layer (network)

- OpenAlex, arXiv (API + OAI-PMH), Crossref, Unpaywall, Europe PMC, DataCite; Semantic Scholar, CORE, PubMed with free keys
- Identity: DOI > OpenAlex > arXiv > S2 > title+year; union-find dedup across sources
- HEAD-probed open-access PDF download, per-paper timeout, status per record
- Every response cached; polite user-agent and per-source delays
- Explicit user action only; OFFLINE=1 disables it entirely
- Hand-off is a folder: data/pdfs/

Local pipeline (offline)

Parse — PyMuPDF4LLM to markdown; header/footer removal; references excised

Chunk — Headings first, then 512-token windows with overlap; section title prepended

Index — SQLite FTS5 + sqlite-vec (NumPy fallback); bge-small 384-d; embedding cache

Retrieve — BM25 | dense | hybrid (weighted RRF) | optional MiniLM cross-encoder rerank

Generate — Ollama Qwen3 on loopback; token-budgeted context; citation-forced prompt; integrity check

Privacy boundary (C1): no module from parse to generate may import an HTTP client; enforced by an AST test on every run. Paper text, chunks, embeddings and answers never leave the machine.

Results: retrieval ablation (30 questions, chunk-level, provisional)

Configuration	Chunk recall@5	Chunk MRR	Retrieval p50
Hybrid (baseline, 512-token chunks)	0.867	0.662	17 ms
BM25 only	0.933	0.588	13 ms
Dense only	0.733	0.458	8 ms
Hybrid + cross-encoder rerank	0.867	0.729	4.1 s
256-token chunks	0.800	0.598	21 ms
1024-token chunks	0.767	0.653	14 ms

What the sweep says

- BM25 wins chunk recall@5 (0.93); dense alone trails (0.73); hybrid sits between and has the best MRR without reranking.
- Reranking lifts MRR to 0.73 but costs ~4 s per query on this CPU (vs 8-21 ms).
- 256-token chunks halve generation time for a small recall loss; 1024-token chunks cost the most and gain nothing.
- Paper-level recall is 1.0 everywhere on 27 papers, which is why the chunk-level metric exists.
- Retrieval latency p50: 8-21 ms; index 12-18 MB; 1,133 chunks.

Results: generation and systems metrics

Metric	Qwen3 1.7B (8 configs)	Qwen3 4B (5 configs)
Citation integrity	1.00 in every config	1.00 in every config
Answers with a citation	100%	83-93% (1-4 lost to the 1,536-token reasoning cap)
Generation p50 per answer	50-195 s	62-99 s (thinks first: ~2.5k chars of reasoning)
Tokens per second	8-12 (post-reboot) / 37 (day 1)	8-10
Resident memory (Ollama)	1.8 GB	3.4 GB; machine peak ~4.1 GB (< 6 GB ceiling)
Correctness on 5 sampled questions	3 of 5 (two wrong numbers)	4 of 5

Eval set: 30 agent-drafted questions over 24 papers, flagged unverified; every metric row carries that flag until human verification. Throughput fell 3-4x after a system reset (CPU, memory, thermal, security and engine settings checked normal); reported as a finding.

Pending works

Evaluation

- Verify the 30-question eval set by hand against the quoted passages (SPEC 9.1) and rerun the sweep on verified ground truth.
- Grade answer correctness manually for the 13 generated configs (results/answers_*.jsonl).
- Run the 4B model over the full sweep once machine throughput is back to day-1 levels; add the Q8 quantisation axis (model not yet pulled).

Corpus, report, demo

- Top up the corpus with retrieval-methods papers once arXiv search is reachable (currently rate-limited from this IP).
- Write the report: architecture, ablation table, limitations, throughput variance.
- Rehearse the Reading Room demo end to end; exercise the 'Add papers' flow live.
- Optional: user feedback from two or three researchers on the evidence panel.

Timeline to complete the remaining works

Week 1 (17-23 Sep)	Verify eval set; grade answers; rerun retrieval sweep on verified questions
Week 2 (24-30 Sep)	Full 4B sweep on a fast day; Q8 axis; arXiv top-up if reachable
Week 3 (1-7 Oct)	Report draft: architecture, ablation table, limitations; figures from runs.csv
Week 4 (8-14 Oct)	Demo rehearsal, README pass, final commit and tag

Effort is bounded by machine time: a full generation sweep with the 4B model takes 6-8 hours on this CPU.

Plan for publication

- No publication is planned for this work.
- Deliverables: public repository (MIT), written report, live demo and results/runs.csv.
- github.com/Charles2524/Research_RAG